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Problem Number – 641

Problem Name – Design Circular Deque

Problem level – Medium

```
class MyCircularDeque {
private:
    vector<int> dq;
    int front, rear, size, capacity;

public:
    MyCircularDeque(int k) {
        capacity = k;
        dq.resize(k);
        front = 0;
        rear = -1;
        size = 0;
    }

    bool insertFront(int value) {
        if (isFull()) return false;

        front = (front - 1 + capacity) % capacity;
        dq[front] = value;

        if (size == 0) rear = front;

        size++;
        return true;
```

```
}

bool insertLast(int value) {
    if (isFull()) return false;

    rear = (rear + 1) % capacity;
    dq[rear] = value;

    if (size == 0) front = rear;

    size++;
    return true;
}

bool deleteFront() {
    if (isEmpty()) return false;

    front = (front + 1) % capacity;
    size--;

    return true;
}

bool deleteLast() {
    if (isEmpty()) return false;

    rear = (rear - 1 + capacity) % capacity;
    size--;

    return true;
}
```

```
    int getFront() {
        if (isEmpty()) return -1;
        return dq[front];
    }

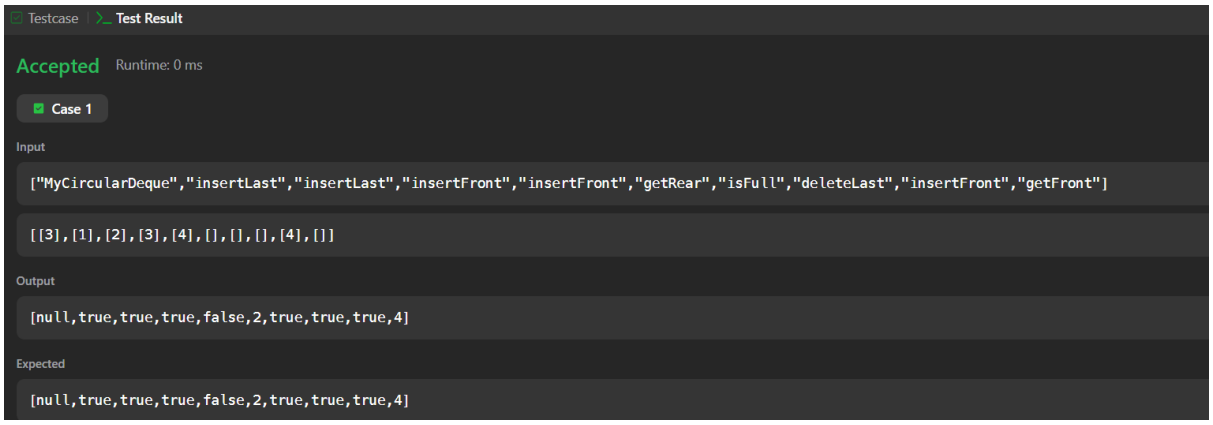
    int getRear() {
        if (isEmpty()) return -1;
        return dq[rear];
    }

    bool isEmpty() {
        return size == 0;
    }

    bool isFull() {
        return size == capacity;
    }
};

/**
 * Your MyCircularDeque object will be instantiated and called as such:
 * MyCircularDeque* obj = new MyCircularDeque(k);
 * bool param_1 = obj->insertFront(value);
 * bool param_2 = obj->insertLast(value);
 * bool param_3 = obj->deleteFront();
 * bool param_4 = obj->deleteLast();
 * int param_5 = obj->getFront();
 * int param_6 = obj->getRear();
 * bool param_7 = obj->isEmpty();
 * bool param_8 = obj->isFull();
```

*/


Testcase
Test Result
Accepted
Runtime: 0 ms
Case 1
Input
["MyCircularDeque","insertLast","insertLast","insertFront","insertFront","getRear","isFull","deleteLast","insertFront","getFront"]
[[3],[1],[2],[3],[4],[],[],[],[4],[]]
Output
[null,true,true,true,false,2,true,true,true,4]
Expected
[null,true,true,true,false,2,true,true,true,4]